Transition-Aware Closed-Loop Restoration for Mobility-Coupled Distribution Systems with Measured Charging and Executable Actions

Weian Guo, Jun Min, and Wuzhao Li

***Abstract*—Restoring mobility-coupled distribution systems requires charging, communication, and repair actions that remain executable under cross-domain degradation. We propose transition-aware physics-guided coupled (PC) rollout, which evaluates multistep threat propagation under a calibrated transition and realizes repair only through integer crew-completion events. Forecast-matched, central PC, and a worst-set robust extension share forecasts and a finite route portfolio; all six policies share packet, SMART-DS/OpenDSS, station-flow, and crew-execution gates. Evaluation combines 148,136 measured Boulder charging transactions, EAGLE-I outages, and a separate SMART-DS feeder. Across 4,096 paired scenarios, central PC reduced dimensionless cost by 0.6327% against the forecast-matched no-transition ablation (paired-difference 95% bootstrap interval $[-7.0372, -5.5195]$); reductions ranged from an operationally negligible 0.0020% nominal effect to 1.1296% in OOD compound cases and persisted under objective reweighting. The robust extension was 0.1708% worse in mean cost and improved neither CVaR nor maximum cost, so no robustness advantage is claimed. Under $20\times$ electrical stress, projection corrected 1,089 raw-infeasible hours while retaining 99.26% of requested service. Theory covers the continuous predictive layer and a conditional finite-route bound, with discrete execution treated separately. The result is a reproducible public-data service-restoration benchmark, not a field deployment or feeder-switching model.**

***Index Terms*—Smart grid restoration, electric vehicle charging, distribution systems, transition-aware rollout, communication networks, crew routing.**

I. Introduction

Distribution grids are becoming coupled operating systems. Electric vehicles create charging demand whose location follows urban activity and road network access. Charging demand consumes feeder margin and may create local backlog when service capacity is insufficient. Communication service supports charging coordination, sensing, and control. A power disturbance can reduce that service when local routers, charging controllers, or edge devices exhaust their backup supply, thereby increasing backlog and affecting mobility operation before the full spatial consequence is observed. Dedicated utility communication and backup batteries can weaken or remove this dependency; power-to-communication coupling is therefore a scenario coefficient, not a universal property. The same effect may move in the opposite direction when mobility congestion changes charging arrival patterns and raises the load on communication-assisted charging services [1], [2]. This coupling changes restoration from a single-layer repair task into a finite-horizon decision problem.

Power and transportation coupling has been studied from planning, pricing, and traffic assignment perspectives. Coordinated charging station planning and traffic power flow models describe how electric mobility reshapes distribution operation [3], [4]. More recent work has linked route choice, charging navigation, and grid states in coupled urban systems [5], [6]. These studies show that charging load cannot be treated as a static exogenous demand when the transportation layer is under stress. At the same time, cyber-physical resilience studies show that service disruption may spread across power, communication, and transportation layers rather than staying inside the first failed domain [7], [8]. A restoration method for such systems must therefore reason about both demand movement and threat propagation.

Forecasting alone is not enough for this restoration task. Graph forecasting models can estimate the near future distribution of mobility demand by using spatial dependence and recent temporal patterns [9], [10]. Digital twin models can evaluate operating constraints and service loss in a form that is closer to the control objective [11], [12]. The missing link is an online decision rule that converts forecasted charging states and propagating cyber-physical threats into restoration actions. A greedy rule that repairs zones with the largest current loss may ignore zones that are not yet highly damaged but are positioned on a future cascade path. A neural rollout rule that follows predicted demand can also miss this effect if restoration is still tied to the current threat map.

We develop transition-aware physics-guided coupled rollout (PC rollout) for this decision task. A graph-recurrent model forecasts measured station arrivals and charging energy. A nonnegative transition propagates road/mobility, power, and communication service degradation, while restoration changes its next state. Coefficients estimated from Boulder charging and EAGLE-I outage records are separated from constructed and sensitivity-only coefficients. The primary central-PC policy scores each candidate under the calibrated transition; a separate robust extension takes the minimum benefit over a declared 21-matrix set. The requested action is then made executable: an independent SMART-DS/OpenDSS loop projects charging power, a packet emulator derates late control commands, and an integer vehicle-routing solver creates work orders and completion events. The next observation closes the

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receding-horizon loop.

The paper makes the following bounded theoretical and algorithmic contributions.

- It formulates an action-dependent, transition-aware finite-horizon rollout under a calibrated central propagation model and separately audits a worst-set extension over data-derived and sign-constrained intervals.
- It turns charging, communication, and repair requests into executable actions through AC feeder projection, packet-deadline feedback, cross-station flow, and route-aware integer crew scheduling.
- It proves conditional finite-horizon sensitivity, finite-portfolio route regret, budget preservation, backlog boundedness, and projected-state invariance, then maps each property to source-level numerical checks.
- It evaluates the complete loop with measured Boulder charging sessions, held-out EAGLE-I outage evidence, SMART-DS power flow, and 4,096 paired uncertain transition scenarios against an information-matched no-transition baseline and the worst-set extension.

The rest of this paper is organized as follows. Section II reviews the related literature. Section III gives the coupled decision model. Section IV presents the propagation model, the rollout policy, and the proof. Section V describes the implementation, experimental design, and empirical results. Section VI discusses scope and limitations, Section VII gives data and code availability, and Section VIII concludes the paper.

II. Related Literature

A. Electric mobility and distribution grids

Electric mobility couples urban travel activity with distribution grid operation. Early charging deployment and demand models linked station access, charging service, and grid impacts [13]–[15]. These models are useful because they connect trip patterns with electricity demand, but most of them address planning or aggregate load estimation. Later studies considered uncertainty in charging station planning, coordinated urban power and transportation networks, and extreme fast charging facilities [4], [16], [17]. Traffic power flow formulations further showed that transportation decisions and electrical constraints can be optimized in a common mathematical model [3], [18].

Recent smart grid studies have moved from planning toward operational coupling. Pricing and navigation models coordinate electric vehicle route choice with charging and grid states [1], [2], [5]. Distributed electric vehicle assignment and charging navigation have also been studied in cyber-physical settings [6]. Robust planning for charging infrastructure has been used to improve resilience in coupled power distribution and transportation systems [8]. Those studies motivate the coupled decision model used here. The difference is the decision time scale. Planning models decide where capacity should be built, while restoration must decide where scarce service and repair effort should move during an unfolding event. In this setting, feeder margin, communication service, charging backlog, and mobility delay must be evaluated together over a finite-horizon.

B. Cyber physical resilience and digital twins

Interdependent infrastructure theory has long recognized that failures can move across network layers. Foundational studies described infrastructure interdependencies and reviewed simulation models for interdependent critical systems [19], [20]. Network science then showed that coupled networks may experience abrupt cascade effects [21]. Spreading process analysis provides mathematical tools for bounding the growth of such processes on graphs [22]. Power system resilience studies connect these ideas to restoration, service continuity, and operation under disruptive events [7], [23], [24].

Physics-constrained security research addresses a related but distinct question: whether adversarial inputs to learned grid functions satisfy power balance, operating limits, and stealth constraints. Formal robustness evaluation has been developed for intelligent security assessment and tree-ensemble smart-grid applications [25], [26]; recent work also constructs physically admissible attacks and defenses for hybrid learned and classical load-frequency control [27]. These studies show why cyber scenarios must respect physical restrictions. They do not, however, allocate executable restoration resources across mobility, communication, and distribution-service states. PC rollout addresses that decision layer, while the independent feeder below constrains action execution without entering the policy score.

Digital twins provide a modeling basis for cyber-physical decision support. General digital twin studies define the connection between a physical asset, a virtual representation, and a data exchange process [11], [12]. Work in networked sensing and industrial informatics emphasizes that the digital representation should remain operationally meaningful rather than only visually descriptive [28]. In the present setting, a useful twin must represent the variables that constrain grid action. These variables include charging service, backlog, feeder margin, communication capacity, and restoration resources.

The restoration problem considered here draws from both lines of work. The threat state is not only an outage indicator. It is a vector field over mobility, power, and communication domains, and it evolves through spatial spread and cross domain couplings. This representation follows the cascade view from interdependent infrastructure analysis, while the cost evaluation uses service variables that can be interpreted by grid operators. The digital twin is therefore not used as a visualization device. It acts as the finite horizon simulator in which restoration actions are tested before the first action is applied.

C. Graph forecasting for charging states

The distribution grid observes future electric mobility demand through forecasts because charging load is driven by travel activity. Diffusion convolutional recurrent networks and spatiotemporal graph convolutional networks established influential approaches for modeling directed traffic dependence [9], [29]. Graph WaveNet and adaptive graph recurrent models improved spatial adaptation and temporal receptive fields [10], [30]. These representative methods justify a compact graph-recurrent state estimator. The forecast architecture is not the contribution evaluated here; the scientific question is how a shared forecast and current observation are converted into a restoration action.

These forecasting methods mainly optimize prediction error. Restoration adds a different requirement. Prediction errors matter through their effect on action selection, and a forecast that is accurate on average can still be insufficient when a future cascade changes the value of restoration at particular zones. For this reason, the predictor in this paper is intentionally kept as a state estimator rather than treated as the main decision maker. The rollout policy uses the forecast to estimate future charging load, then uses the propagation model to form the cascade envelope that drives restoration. This distinction separates demand estimation from cyber-physical risk allocation, which is central to the method and to the experimental comparisons.

III. Coupled Smart Grid Operation Model

A. Operating state

Let $G = (V, E)$ be an urban charging service graph with N zones. The weighted adjacency matrix W is obtained from origin and destination flows that describe where electric mobility demand is likely to move. A symmetrized normalized matrix is denoted by A. At decision time t, the measured and estimated state is

$$x_t = (m_t, e_t, q_t, c_t, p_t, z_t), \tag{1}$$

where $m_t \in \mathbb{R}_+^N$ is mobility activity that drives charging demand, $e_t \in \mathbb{R}_+^N$ is charging energy demand, $q_t \in \mathbb{R}_+^N$ is charging backlog, $c_t \in \mathbb{R}_+^N$ is communication load, $p_t \in \mathbb{R}_+^N$ is feeder load, and $z_t = (z_t^r, z_t^p, z_t^c)$ is the mobility, power, and communication threat field. Each threat component lies in $[0, 1]^N$. It is a normalized service-degradation intensity, not the binary state of one breaker, line, or router. For example, $z_i^p = 0.6$ means a 60% derating in the zone-level power-dependent service factor; $z_i^r = 0.6$ produces a 60% proportional mobility impairment before the communication term. A binary outage can be represented as a high-intensity limiting case. The scenario scope includes cyber-induced loss of sensing/control availability and physical or service failures; the model does not identify attack type.

The requested decision is

$$a_t = (u_t^{\mathrm{ch}}, u_t^{\mathrm{com}}, u_t^{\mathrm{res}}). \tag{2}$$

The first component allocates charging service capacity, the second allocates communication service capacity, and the third is a continuous restoration priority used by the predictive scoring layer. It is not itself a completed repair. The feasible set is

$$\begin{aligned} \mathcal{A} = \{a_t \geq 0 \,|\, &\mathbf{1}^\top u_t^{\mathrm{ch}} \leq B_{\mathrm{ch}}, \\ &\mathbf{1}^\top u_t^{\mathrm{com}} \leq B_{\mathrm{com}}, \\ &\mathbf{1}^\top u_t^{\mathrm{res}} \leq B_{\mathrm{res}}\}. \end{aligned} \tag{3}$$

Here B_{ch}, B_{com}, and B_{res} are the available charging, communication, and normalized restoration resources during one decision interval. The continuous restoration variable is a service-level priority used to form candidate work orders. Section V converts these candidates into integer crews, explicit road routes, service times, and repair-completion events. At every policy hour, charging action is also projected through the SMART-DS/OpenDSS feasibility loop before service is credited and backlog is updated.

B. Grid and service equations

The continuous prediction layer uses monotone service equations that are continuous, piecewise smooth, and Lipschitz on the stated bounded domain. The minimum, positive-part, and clipping operators need not be differentiable at their switching points. For zone i and rollout step h, available communication service is

$$\bar{c}_{i,h} = u_{i,h}^{\mathrm{com}}(1 - z_{i,h}^p)(1 - \alpha_c z_{i,h}^c), \tag{4}$$

where $\alpha_c \in [0, 1]$ measures direct communication degradation. The load placed on communication is

$$c_{i,h} = \beta_m m_{i,h} + \beta_e e_{i,h} + \beta_r z_{i,h}^r + \beta_p z_{i,h}^p. \tag{5}$$

The resulting communication loss is

$$\ell_{i,h}^c = [c_{i,h} - \bar{c}_{i,h}]_+. \tag{6}$$

Communication support affects charging coordination and mobility service through

$$\eta_{i,h}^c = 1 - \min\left\{0.9, \frac{\ell_{i,h}^c}{1 + c_{i,h}}\right\}. \tag{7}$$

This algebraic relation is an hourly workload model. Executability is supplied by a finite-buffer packet emulator with retransmission and deadline logic. If $d_{i,h} \in [0, 1]$ is the simulated fraction of control packets delivered before their deadline, the automated action applied to the service and threat transitions is

$$\tilde{a}_{i,h} = d_{i,h} a_{i,h}. \tag{8}$$

The emulator keeps full service rate while backup power is available and applies power-induced service derating only after backup exhaustion. Packet loss and delay therefore change the implemented action rather than appearing only as an after-the-fact diagnostic. Before charging service is credited, the packet-derated request is mapped to three-phase SMART-DS loads and scaled by the largest AC-feasible fraction $\alpha_h^{\mathrm{AC}} \in [0, 1]$ found by bisection,

$$u_{i,h}^{\mathrm{ch,exec}} = \alpha_h^{\mathrm{AC}} \tilde{u}_{i,h}^{\mathrm{ch}}. \tag{9}$$

The feeder must converge, all energized-node voltages must lie in $[0.95, 1.05]$ p.u., and maximum line loading must not exceed 1.0 p.u. Charging service is therefore

$$s_{i,h}^{\mathrm{ch}} = \min\left\{e_{i,h} + \rho_q q_{i,h}, u_{i,h}^{\mathrm{ch,exec}}(1 - z_{i,h}^p)\eta_{i,h}^c\right\}. \tag{10}$$

The charging backlog evolves as

$$q_{i,h+1} = \left[e_{i,h} + \rho_q q_{i,h} - s_{i,h}^{\mathrm{ch}}\right]_+. \tag{11}$$

We impose $0 \leq \rho_q < 1$ (the implementation uses $\rho_q = 0.22$). If hourly arrival energy is bounded by $\bar{e}$, then $q_{i,h+1} \leq \bar{e} + \rho_q q_{i,h}$ and consequently $q_{i,h} \leq \max\{q_{i,0}, \bar{e}/(1 - \rho_q)\}$.

The original zone-level feeder-margin surrogate is retained only as an interpretable service diagnostic,

$$\ell_{i,h}^{\mathrm{fm}} = \left[p_i^0 + s_{i,h}^{\mathrm{ch}} - \bar{p}_i(1-\alpha_p z_{i,h}^p)\right]_+, \tag{12}$$

where p_i^0 is an abstract base load and $\bar{p}_i$ is an abstract local capacity. It is neither called nor used as a voltage violation: executable charging is determined by the online OpenDSS projection in (9). Mobility delay is

$$\ell_{i,h}^m = m_{i,h}\left(z_{i,h}^r + \rho_c(1-\eta_{i,h}^c)\right). \tag{13}$$

Unserved charging is defined once as $\ell_{i,h}^e = [e_{i,h} + \rho_q q_{i,h} - s_{i,h}^{\mathrm{ch}}]_+$, and power-dependent service loss is $\ell_{i,h}^p = z_{i,h}^p(e_{i,h} + \rho_q q_{i,h})$. The coefficients in these equations have fixed physical roles. The parameters β_m, β_e, β_r, and β_p convert mobility activity, charging demand, mobility threat, and power threat into communication workload. The coefficient ρ_q is backlog carryover, and α_p is the abstract capacity derating caused by power threat. The coefficient ρ_c converts communication service reduction into delayed-trip equivalents. The stage risk is

$$\begin{aligned} L(x_h, a_h) = \sum_{i=1}^{N} \big(&\lambda_m \ell_{i,h}^m + \lambda_e \ell_{i,h}^e \\ &+ \lambda_c \ell_{i,h}^c + \lambda_p \ell_{i,h}^p \\ &+ \lambda_z[z_{i,h}^p z_{i,h}^c + z_{i,h}^r(1-\eta_{i,h}^c)]\big). \end{aligned} \tag{14}$$

The nonnegative weights λ_m, λ_e, λ_c, λ_p, and λ_z set the relative contributions of mobility delay, unmet charging demand, communication loss, power-dependent service loss, and cross-domain exposure. The objective is dimensionless because these heterogeneous terms are combined with fixed relative weights; it is not a monetary operating cost. Unserved charging appears only through ℓ^e. All coefficients are held fixed across policies.

For a feasible policy π, the finite-horizon rollout objective at decision time t is

$$J_H^\pi(x_t, \zeta_t) = \sum_{h=0}^{H-1} \gamma^h L(x_{t+h}, a_{t+h}), \tag{15}$$

$$a_{t+h} = \pi(x_{t+h}, \zeta_{t+h}, A). \tag{16}$$

The closed loop controller applies only the first action from the horizon calculation. At the next decision time, the observed state replaces the previous prediction and the calculation is repeated. In this receding horizon interpretation, the next state measurement absorbs forecast errors and threat disturbances before the following decision.

C. Forecasting component

The forecaster estimates mobility and charging demand.

$$\hat{y}_{t+1:t+H} = f_\theta(y_{t-L+1:t}, A), \tag{17}$$

where $y_t = (m_t, e_t)$, L is the historical window, and H is the rollout horizon. The implemented model first mixes node features through A, then updates zone states using a recurrent cell, and finally maps the hidden state to the horizon. Keeping the forecaster small makes its role clear inside the larger decision model. Classical baselines and the learned model are compared in Section V so that the decision results can be interpreted relative to the forecasting quality.

The input series is transformed by a logarithmic map and normalized using only the training portion of the calendar year. For a time index τ,

$$\tilde{y}_\tau = \frac{\log(1+y_\tau) - \mu}{\sigma}. \tag{18}$$

The graph recurrent forecaster uses one graph mixing operation before each recurrent update. With hidden state h_ℓ inside the historical window,

$$g_\ell = A\tilde{y}_{t-L+\ell}, \tag{19}$$

$$h_\ell = \mathrm{GRU}(W_g g_\ell, Ah_{\ell-1}), \tag{20}$$

$$\hat{y}_{t+1:t+H} = W_o\,\phi(h_L). \tag{21}$$

Here ϕ denotes the output transformation used by the prediction head. The primary forecaster is trained with the Huber loss

$$\mathcal{L}_{\mathrm{for}} = \mathcal{L}_{\mathrm{Huber}}(\hat{y}, y). \tag{22}$$

Its spatial inductive bias comes from the fixed origin–destination graph mixing inside the recurrent architecture, not from an additional physical penalty. Three random seeds are trained, and the checkpoint is selected only by the validation score defined in the accompanying code. An auxiliary graph-smooth training mode is retained in the archive for reproducibility, but it is not used by the revised rollout evaluation because its validation score is worse.

IV. Physics Guided Smart Grid Rollout

A. Threat propagation

Let $\zeta_h = (z_h^r, z_h^p, z_h^c) \in [0,1]^{3N}$. Two transitions are kept explicit because prediction and execution have different roles. The continuous transition used only to score a requested restoration priority is

$$\hat{\zeta}_{h+1} = \Pi_{\mathcal{Z}}\left(M\hat{\zeta}_h - Ru_h^{\mathrm{pri}} + \hat{\xi}_h\right), \tag{23}$$

where $\mathcal{Z} = [0, 0.98]^{3N}$, $\Pi_{\mathcal{Z}}$ is Euclidean projection, $M \geq 0$ is the cross-domain propagation matrix, $R \geq 0$ maps the zone-level priority u_h^{pri} to predicted road, power, and communication relief, and $\hat{\xi}_h$ is the innovation assumed by the scorer. Equation (23) does not update the realized state.

For execution, let $\bar{r}_{i,h} \in \{0,1\}$ indicate that a dispatched work order in zone i has completed by the end of interval h, and define $D(\bar{r}_h) = I_3 \otimes \mathrm{diag}(1-\bar{r}_h)$. The main simulator uses

$$\zeta_{h+1} = D(\bar{r}_h)\Pi_{\mathcal{Z}}\left(M\zeta_h + \xi_h\right). \tag{24}$$

Waiting, travel, and service do not change $\bar{r}_h$; only a logged crew completion event changes it from zero to one. Thus no continuous priority is credited as realized repair. With $K_d = \rho_d I + \sigma A$ for $d \in \{r,p,c\}$, the implemented matrix is

$$M = \begin{bmatrix} K_r & \tau_{pr}I & \tau_{cr}I \\ 0 & K_p & \tau_{cp}A \\ \tau_{rc}A & \tau_{pc}I & K_c \end{bmatrix}. \tag{25}$$

The normalized adjacency A enters both within-domain spread and selected cross-domain paths. Boulder charging deficits estimate ρ_r and σ; EAGLE-I Boulder County outages estimate ρ_p; aligned outage and charging records bound τ_{pr}; and the packet emulator constructs the power-to-communication interval for τ_{pc}. Coefficients lacking joint field observations are explicitly labeled sensitivity-only. Bootstrap and sign-constrained intervals define an uncertainty set $\mathcal{U}$. The central matrix has estimated spectral radius 1.028, so no asymptotic contraction is claimed; projection makes the implemented state set invariant and the policy is evaluated over a finite six-hour horizon and a 100,000-step stress test.

PC rollout scores restoration by its action-dependent marginal benefit. Let $w_{i,h}$ be a bounded exposure weight formed from the shared mobility and energy forecast, and let $\Omega = \mathrm{diag}(\omega_r, \omega_p, \omega_c)$. For a candidate first restoration vector v, define

$$\mathcal{C}_t(v) = \sum_{h=0}^{H-1} \sum_{i=1}^{N} w_{i,h} \mathbf{1}^\top \Omega \hat{\zeta}_{i,t+h}(v), \tag{26}$$

where $\hat{\zeta}_{t+h}(v)$ follows (23), using v only in the first predicted transition. For reference quantum δ, the marginal benefit of repairing zone i is $b_{i,t}(M) = [\mathcal{C}_t(0; M) - \mathcal{C}_t(\delta e_i; M)]_+$. The primary transition-aware policy uses the calibrated central matrix M_c, $b^{\mathrm{cen}}_{i,t} = b_{i,t}(M_c)$. A separately evaluated robust extension uses the lower envelope

$$b^{\mathrm{rob}}_{i,t} = \min_{M \in \mathcal{U}_V} b_{i,t}(M), \tag{27}$$

where $\mathcal{U}_V$ contains the central matrix, joint lower and upper vertices, and one-coefficient-at-a-time interval endpoints (21 matrices in the implementation). The policy-specific time-indexed restoration score is

$$s^{\mathrm{res}}_{i,t} = \omega_p z^p_{i,t} + \omega_c z^c_{i,t} + \omega_r z^r_{i,t} + \omega_\Gamma b^\pi_{i,t}. \tag{28}$$

Here $b^\pi = b^{\mathrm{cen}}$ for the primary central-PC method and $b^\pi = b^{\mathrm{rob}}$ only for the robust extension. The restoration action uses a small uniform reserve and a concave risk allocation.

$$u^{\mathrm{res}}_i = \frac{\eta B_{\mathrm{res}}}{N} + (1-\eta) B_{\mathrm{res}} \frac{\sqrt{s^{\mathrm{res}}_i + \epsilon}}{\sum_{j=1}^{N} \sqrt{s^{\mathrm{res}}_j + \epsilon}}, \tag{29}$$

where η is the reserve fraction. The square root creates concavity. It prevents a single zone from taking nearly all restoration effort while still giving more effort to zones with larger cascade exposure. The weights ω_p, ω_c, ω_r, and ω_Γ are nonnegative constants that control direct domain exposure and the multi-step action benefit. The constant ϵ is a small positive stabilizer that keeps every zone eligible for a nonzero reserve.

Charging and communication capacities are assigned with the same allocator so that all resource decisions share the same feasibility structure. For any budget B, nonnegative score vector v, and reserve fraction η_B, define

$$\Phi_i(B, v, \eta_B) = \frac{\eta_B B}{N} + (1-\eta_B) B \frac{\sqrt{v_i + \epsilon}}{\sum_{j=1}^{N} \sqrt{v_j + \epsilon}}. \tag{30}$$

PC rollout uses

$$u^{\mathrm{ch}}_i = \Phi_i(B_{\mathrm{ch}}, g^{\mathrm{ch}}, \eta_{\mathrm{ch}}), \tag{31}$$
$$u^{\mathrm{com}}_i = \Phi_i(B_{\mathrm{com}}, g^{\mathrm{com}}, \eta_{\mathrm{com}}), \tag{32}$$

where

$$g^{\mathrm{ch}}_{i,t} = \sum_{h=1}^{H} \hat{e}_{i,t+h} + \kappa_q q_{i,t} + \kappa_{\mathrm{ch}} \left(z^r_{i,t} + z^p_{i,t} + z^c_{i,t} \right), \tag{33}$$

$$g^{\mathrm{com}}_{i,t} = \sum_{h=1}^{H} \hat{m}_{i,t+h} + \kappa_E \left(\sum_{h=1}^{H} \hat{e}_{i,t+h} + \kappa_q q_{i,t} \right) + \kappa_{\mathrm{com}} \left(z^r_{i,t} + z^p_{i,t} + z^c_{i,t} \right). \tag{34}$$

Service capacity follows predicted demand, while the requested restoration priority follows the future cascade envelope. Charging and communication resources serve expected load. In the scoring layer, priority perturbs (23); in the execution layer, only the completion indicator in (24) changes realized threat. The coefficients κ_q, κ_{ch}, κ_E, and κ_{com} determine how observed backlog, current threat, and forecasted charging demand alter the time-varying service scores. For the main experiment, an initial threat realization and adjacency-based propagation create a policy-independent set of at most 36 work orders. Four integer crews start from a common depot. Candidate routes are constructed from static, greedy, forecast-matched, central-PC, robust-PC, and uniform priorities using the same jobs, travel matrix, service-time draws, and dispatch uniforms for every policy. Forecast-matched, central PC, and robust PC see exactly this same finite route portfolio. Forecast-matched minimizes current exposure-weighted completion without propagating M; central PC minimizes predicted integrated risk under the calibrated central M; robust PC minimizes the worst predicted integrated risk over the 21-matrix uncertainty set. Travel uses geocoded station locations, 1.25 road circuity, and 30 km/h; service-time scenarios are informed by observed EAGLE-I post-peak recovery times. Because those county traces are not work orders, they are an explicit scenario prior rather than measured crew durations. A dispatched job changes no restoration state while waiting or being serviced. Only its logged completion event clears local threat, after which the zone remains repaired within the horizon. The mobility graph A is used for propagation; crew travel uses the separate geocoded road-distance matrix.

Algorithm 1 states the decision rule. Forecasting, propagation, service evaluation, and restoration allocation are kept as separate calculations so that the effect of each component can be inspected.

Lemma 1. The restoration-priority allocation in (29) is feasible for every nonnegative score vector.

Proof. Each score satisfies $s^{\mathrm{res}}_i \geq 0$ and $\epsilon > 0$, so every square root in (29) is positive. Hence $u^{\mathrm{res}}_i \geq 0$ for all zones. Summing (29) over all zones gives

$$\sum_{i=1}^{N} u^{\mathrm{res}}_i = \sum_{i=1}^{N} \frac{\eta B_{\mathrm{res}}}{N} + (1-\eta) B_{\mathrm{res}} \frac{\sum_{i=1}^{N} \sqrt{s^{\mathrm{res}}_i + \epsilon}}{\sum_{j=1}^{N} \sqrt{s^{\mathrm{res}}_j + \epsilon}} = \eta B_{\mathrm{res}} + (1-\eta) B_{\mathrm{res}} = B_{\mathrm{res}}. \tag{35}$$

The requested priority satisfies the nonnegativity and budget constraints. □

B. Mathematical properties

The Lipschitz results below concern the continuous pre-execution prediction and allocation operators. They exclude

Algorithm 1: Transition-aware physics-guided coupled rollout

Require: Current state x_t, graph A, budgets B_{ch}, B_{com}, B_{res}, horizon H, and trained forecaster f_θ

Ensure: Executable service action and event-driven repair update

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Predict ŷ_{t+1:t+H} with f_θ
Initialize mobility, power, and communication threats from the observed state
for each zone i do
    Roll out (23) at M_c with no first action and candidate δe_i
    Set b^cen_{i,t} ← [C_t(0; M_c) − C_t(δe_i; M_c)]_+
    Optionally evaluate b^rob_{i,t} over U_V for the robust extension
end for
Allocate charging and communication service with (30)
Compute requested restoration priority u^pri_t with (29)
Translate priorities into requested work orders
Apply the packet deadline gate to form dispatched work orders
Construct the common finite route portfolio over dispatched orders
Select a route by current, central-M, or worst-set scoring according to policy
Gate charging and communication actions by on-time packets
Project charging through OpenDSS; credit only the executable service
Advance (24) only when a crew completion event occurs
Observe the next state and repeat
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packet deadline switches, OpenDSS accept/reject tests and bisection, station-flow active-set changes, finite-route argmin selection, and integer completion events. Those executable modules are covered instead by the finite-portfolio bound and constraint-preservation proposition below.

The analysis uses the Euclidean norm. The projection onto a closed convex set is nonexpansive, so

$$\|\Pi_{\mathcal{Z}}(u) - \Pi_{\mathcal{Z}}(v)\| \le \|u - v\|. \tag{36}$$

Condition 1. On the stated bounded continuous prediction domain and for all feasible requested actions, the one-step service-state operator $F_c(x, a, \zeta)$ is Lipschitz in (x, ζ) with finite constants L_x and L_z.

$$\|F_c(x, a, \zeta) - F_c(\tilde{x}, a, \tilde{\zeta})\| \le L_x\|x - \tilde{x}\| + L_z\|\zeta - \tilde{\zeta}\|. \tag{37}$$

Condition 2. The stage risk in (14) is Lipschitz in (x, ζ) with constants C_x and C_z for all feasible actions.

$$|L(x, a, \zeta) - L(\tilde{x}, a, \tilde{\zeta})| \le C_x\|x - \tilde{x}\| + C_z\|\zeta - \tilde{\zeta}\|. \tag{38}$$

These conditions are justified for the continuous operators on the bounded domain, not for all of $\mathbb{R}^n$ or for the complete hybrid executor. Projection, scalar minimum, positive part, and clipping are nonexpansive but need not be differentiable at switching points. The affine maps have finite operator norms, products are Lipschitz on a bounded set, and the backlog is bounded by $\rho_q < 1$. Finite composition therefore implies existence of finite constants for this continuous layer. Numerical property tests check invariants and sampled contracts; they do not estimate or prove global Lipschitz constants.

Lemma 2. For fixed action sequence and disturbance sequence, if two rollout trajectories start from threat fields ζ_0 and $\tilde{\zeta}_0$ and from the same physical state, then

$$\|\zeta_h - \tilde{\zeta}_h\| \le \|M\|^h\|\zeta_0 - \tilde{\zeta}_0\|. \tag{39}$$

Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. Because the action and innovation sequences are shared, (23) and the nonexpansiveness in (36) give $\Delta_{h+1} \le \|M\|\Delta_h$. Iteration proves (39). □

Lemma 3. Under Condition 1, the service-state error satisfies

$$E_h \le \sum_{k=0}^{h-1} L_x^{h-1-k} L_z \Delta_k. \tag{40}$$

Proof. The two rollouts use the same action sequence. Condition 1 gives $E_{h+1} \le L_x E_h + L_z\Delta_h$. Since the initial physical state is the same, $E_0 = 0$. Expanding the recursion gives $E_1 \le L_z\Delta_0$ and $E_2 \le L_x L_z \Delta_0 + L_z\Delta_1$. Repeating the expansion gives (40). □

Theorem 1. Let $J_H(x_0, \zeta_0)$ be the discounted H step risk of a fixed feasible action sequence, and let $\tilde{J}_H(x_0, \tilde{\zeta}_0)$ be the corresponding risk for an initial threat estimate $\tilde{\zeta}_0$. For any $\alpha > 0$, define

$$\chi_\alpha = \max\{\|M\| + \alpha L_z,\ L_x\}. \tag{41}$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \le \varepsilon$, then

$$|J_H(x_0, \zeta_0) - \tilde{J}_H(x_0, \tilde{\zeta}_0)| \le \varepsilon\left(C_z + \frac{C_x}{\alpha}\right)\sum_{h=0}^{H-1}(\gamma\chi_\alpha)^h. \tag{42}$$

For every finite H the sum is finite. If $\gamma\chi_\alpha < 1$, it is also uniformly bounded as the horizon increases by

$$\varepsilon\left(C_z + \frac{C_x}{\alpha}\right)\frac{1}{1 - \gamma\chi_\alpha}. \tag{43}$$

Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. Lemma 2 and Condition 1 give the one-step bounds

$$\Delta_{h+1} \le \|M\|\Delta_h, \tag{44}$$

$$E_{h+1} \le L_z\Delta_h + L_x E_h. \tag{45}$$

For a fixed $\alpha > 0$, define the weighted error $V_h = \Delta_h + \alpha E_h$. Multiplying (45) by α and adding (44) gives

$$\begin{aligned} V_{h+1} &\le (\|M\| + \alpha L_z)\Delta_h + \alpha L_x E_h \\ &\le \chi_\alpha\Delta_h + \alpha\chi_\alpha E_h = \chi_\alpha V_h. \end{aligned} \tag{46}$$

The two rollouts start from the same physical state, so $E_0 = 0$ and $V_0 = \Delta_0 \le \varepsilon$. Induction gives $V_h \le \chi_\alpha^h\varepsilon$ for every h. Since $\Delta_h \le V_h$ and $E_h \le V_h/\alpha$, Condition 2 yields

$$|L(x_h, a_h, \zeta_h) - L(\tilde{x}_h, a_h, \tilde{\zeta}_h)| \le \left(C_z + \frac{C_x}{\alpha}\right)\chi_\alpha^h\varepsilon. \tag{47}$$

Multiplying by γ^h and summing from $h = 0$ to $H - 1$ gives (42). The infinite geometric series gives the uniform bound when $\gamma\chi_\alpha < 1$. □

Condition 3. The continuous service transition, stage risk, and pre-execution PC allocation map π_c are Lipschitz in the action argument on the compact feasible set. There are finite constants L_a, C_a, L_π^x, and L_π^z such that

$$\|F_c(x, a, \zeta) - F_c(x, \tilde{a}, \zeta)\| \le L_a \|a - \tilde{a}\|, \tag{48}$$

$$|L(x, a, \zeta) - L(x, \tilde{a}, \zeta)| \le C_a \|a - \tilde{a}\|, \tag{49}$$

$$\|\pi_c(x, \zeta, A) - \pi_c(\tilde{x}, \tilde{\zeta}, A)\| \le L_\pi^x \|x - \tilde{x}\| + L_\pi^z \|\zeta - \tilde{\zeta}\|. \tag{50}$$

The last inequality concerns the continuous score-to-allocation map: candidate rollouts use (23), positive part is nonexpansive, and the allocator uses $\sqrt{s + \epsilon}$ with $\epsilon > 0$. Its derivative is bounded by $1/(2\sqrt{\epsilon})$, and the positive normalization denominator is bounded away from zero. This supplies a finite constant on the compact continuous domain. It does not apply to the subsequent finite-route argmin, packet events, OpenDSS projection switch, or integer crew completion process.

Corollary 1 (continuous pre-execution sensitivity). Let $J_H^{\pi_c}(x_0, \zeta_0)$ and $\tilde{J}_H^{\pi_c}(x_0, \tilde{\zeta}_0)$ be the predictive risks generated by the continuous PC score-and-allocation layer from the same physical state and two initial threat estimates. Define

$$\bar{L}_x = L_x + L_a L_\pi^x, \quad \bar{L}_z = L_z + L_a L_\pi^z, \tag{51}$$

$$\bar{C}_x = C_x + C_a L_\pi^x, \quad \bar{C}_z = C_z + C_a L_\pi^z. \tag{52}$$

For any $\alpha > 0$, let

$$\bar{\chi}_\alpha = \max\left\{\|M\| + \|R\| L_\pi^z + \alpha \bar{L}_z,\ \bar{L}_x + \frac{\|R\| L_\pi^x}{\alpha}\right\}. \tag{53}$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \le \varepsilon$, then

$$|J_H^{\pi_c}(x_0, \zeta_0) - \tilde{J}_H^{\pi_c}(x_0, \tilde{\zeta}_0)| \le \varepsilon\left(\bar{C}_z + \frac{\bar{C}_x}{\alpha}\right) \sum_{h=0}^{H-1} (\gamma \bar{\chi}_\alpha)^h. \tag{54}$$

Proof. Let $a_h = \pi_c(x_h, \zeta_h, A)$ and $\tilde{a}_h = \pi_c(\tilde{x}_h, \tilde{\zeta}_h, A)$. Condition 3 gives

$$\|a_h - \tilde{a}_h\| \le L_\pi^x E_h + L_\pi^z \Delta_h.$$

Combining this bound with Conditions 1 and 3 and the restoration term in (23) yields

$$E_{h+1} \le \bar{L}_x E_h + \bar{L}_z \Delta_h, \tag{55}$$

$$\Delta_{h+1} \le (\|M\| + \|R\| L_\pi^z) \Delta_h + \|R\| L_\pi^x E_h. \tag{56}$$

For $V_h = \Delta_h + \alpha E_h$, the preceding two inequalities give $V_{h+1} \le \bar{\chi}_\alpha V_h$. Since $V_0 \le \varepsilon$, induction gives $V_h \le \bar{\chi}_\alpha^h \varepsilon$. The stage risk difference satisfies

$$|L(x_h, a_h, \zeta_h) - L(\tilde{x}_h, \tilde{a}_h, \tilde{\zeta}_h)| \le \bar{C}_x E_h + \bar{C}_z \Delta_h.$$

Using $E_h \le V_h/\alpha$ and $\Delta_h \le V_h$, then summing the discounted stage differences, gives the stated bound. □

Theorem 2 (conditional finite-portfolio regret). Let $\mathcal{P}$ be the finite portfolio of integer crew routes available at one decision time. Let $Q(P)$ be the realized integrated risk of route P and $\widehat{Q}(P)$ its information-feasible PC estimate. Suppose $|Q(P) - \widehat{Q}(P)| \le \varepsilon_Q$ for every $P \in \mathcal{P}$, and the route solver returns $\widehat{P}$ satisfying

$$\widehat{Q}(\widehat{P}) \le \min_{P \in \mathcal{P}} \widehat{Q}(P) + \delta_s. \tag{57}$$

Then

$$Q(\widehat{P}) \le \min_{P \in \mathcal{P}} Q(P) + 2\varepsilon_Q + \delta_s. \tag{58}$$

If the realized optimum P^* is unique with objective gap $\Delta_Q > 2\varepsilon_Q + \delta_s$, then $\widehat{P} = P^*$.

Proof. Let $P^* = \arg\min_P Q(P)$. Uniform approximation and solver tolerance give $Q(\widehat{P}) \le \widehat{Q}(\widehat{P}) + \varepsilon_Q \le \widehat{Q}(P^*) + \delta_s + \varepsilon_Q \le Q(P^*) + 2\varepsilon_Q + \delta_s$. For any $P \ne P^*$, the gap condition implies $\widehat{Q}(P) > \widehat{Q}(P^*) + \delta_s$, so no such plan can be returned. □

Proposition 1 (implementation feasibility and invariance). Assume the base feeder operating point is feasible, packet enactment satisfies $0 \le d_{i,h} \le 1$, service arrivals are bounded, $0 \le \rho_q < 1$, and every accepted work-order instance has a finite integer route. Then every implemented step satisfies the three resource budgets, the packet-derated action cannot violate a budget that its requested action satisfied, the OpenDSS action projection returns an AC-feasible charging action, backlog remains bounded, and $\zeta_h \in \mathcal{Z}$ for all h.

Proof. Lemma 1 applies to each allocator. Elementwise multiplication by $d_h \in [0, 1]^N$ preserves nonnegativity and cannot increase any budget sum. The feeder adapter searches decreasing charging fractions and includes zero incremental EV action; feasibility of the base point therefore supplies a feasible fallback. The backlog bound follows from the stated scalar recursion, and projection in (24) makes $\mathcal{Z}$ invariant. Finite route coverage makes every accepted repair completion well-defined. The implementation checks these claims using 10,000 random budget tests, 5,000 integer-crew tests, all 8,117 nonzero 2023 test hours for the backlog bound, 100,000 projected threat steps, 14,902 standalone SMART-DS AC scenarios, 147,456 online primary policy-hours, 36,864 online stress policy-hours, and explicit route-completion, route-coverage, packet-backup, and station-flow assertions. □

Theorem 1 and Corollary 1 are conditional finite-horizon sensitivity statements for the continuous predictive layer, not for the full hybrid execution loop and not proofs of asymptotic stability or global policy optimality. Theorem 2 is a conditional finite-portfolio bound: we do not claim a numerical guarantee because a uniform empirical estimate of ε_Q is not available. Proposition 1 instead establishes executable constraint preservation under its stated assumptions. Random property checks exercise the covered operators and implementation assertions; they are not a proof of a global Lipschitz constant.

V. Experimental Design and Results

A. Evidence sources, preprocessing, and splits

The revised evaluation separates measured demand, outage context, electrical feasibility, communication enactment, and crew execution instead of deriving all layers from one mobility proxy. Table I records the role and boundary of each source.

The primary demand source is the City of Boulder Electric Vehicle Charging Station Data under a CC0 license [31]. Its metadata defines one row as one transaction and provides local transaction time, station identity, charging duration, and delivered energy. The local archive contains all 148,136 transactions released for 2018–2023, with 50 station records

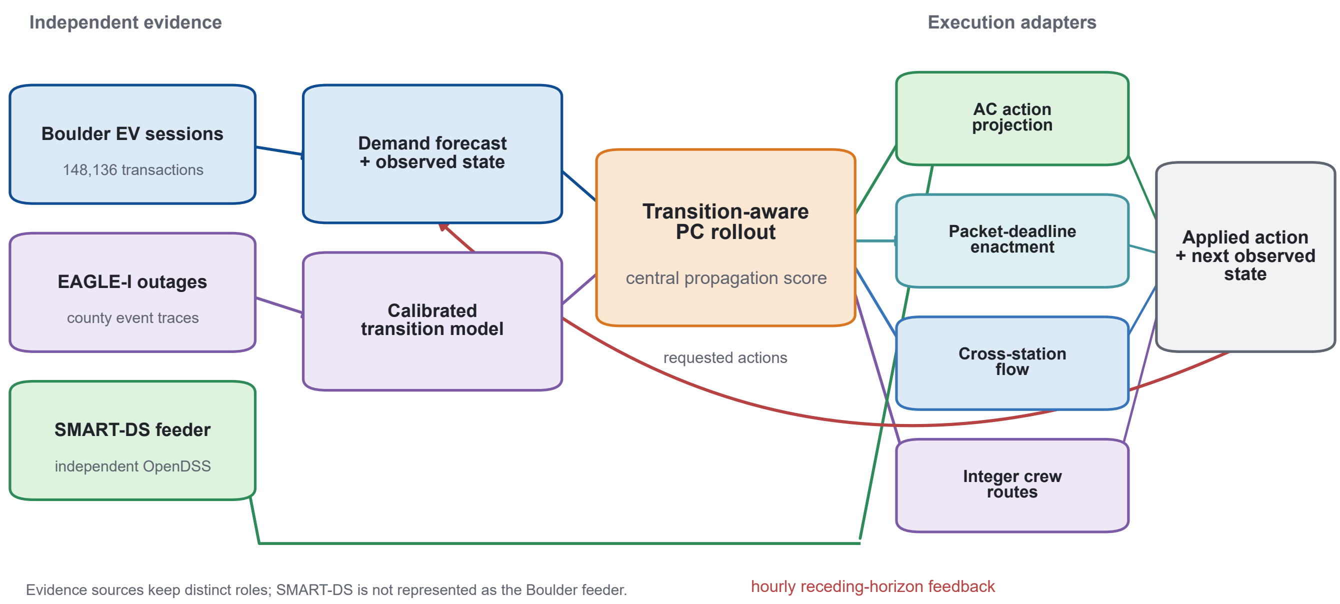

Fig. 1. Executable transition-aware-PC closed loop. Boulder charging sessions supply measured demand and held-out forecasts; EAGLE-I supplies county-level outage calibration; SMART-DS independently constrains feeder action. PC rollout issues requested actions, but service and threat states are updated only after AC projection, packet-deadline enactment, cross-station flow, and integer crew routing. The next observation restarts the hourly horizon. The three public data sources have distinct roles and are not represented as a geographically coincident utility system.

TABLE I
INDEPENDENT EVIDENCE SOURCES AND THEIR NON-INTERCHANGEABLE ROLES

Source	Scale	Observed fields	Role in this study
City of Boulder EV sessions	148,136 transactions; 50 station records; 2018–2023	Arrival time, charging duration, energy (kWh), station/address	Primary measured charging demand; fixed 2018–2021/2022/2023 train/validation/test split
EAGLE-I	249,316,543 national rows streamed; 819,528 retained in Boulder and six adjacent counties; 2014–2025	County, UTC timestamp, customers out	Power-state persistence, event/recovery distributions, and uncertainty bounds; missing rows are not silently coded as zero
SMART-DS SFO P1U feeder	80 buses, 219 nodes, 27 loads, 67 lines, 13 transformers	OpenDSS circuit, equipment and peak-load operating point	Online projection in all 147,456 primary policy-hours plus an integrated stress test; not used to train or score the policy
Packet/route modules	12,000 packet stress rows; 4,096 paired main scenarios; up to 36 jobs and four crews per scenario	Queue events, retries, deadlines, backup; integer crews, travel, service and completion times	Gates requested controls and updates threat state only after a routed repair-completion event

and 1,252,719.654 measured positive kWh. Four records lack an end time, 16,113 report zero energy, no duplicate source identifier or full row was found, and no negative energy was retained. Arrival counts are exact transaction counts by start hour. Session energy is measured; its hourly load profile is a stated proxy obtained by spreading each session's energy uniformly over the reported charging duration. Conservation checks recover all 148,136 arrivals and the measured positive energy to within 0.001 kWh.

Time was split before model fitting: 2018–2021 for training, 2022 for model selection, and 2023 for final testing. No random row split crosses this calendar boundary. A second five-fold spatial test withheld contiguous west–east station blocks from target fitting. In that inductive deployment test, a held-out station's pre-2022 calibration and observed 1, 24, and 168 h lags were permitted, but its target values never entered coefficient fitting.

Power-event evidence came from the CC BY 4.0 EAGLE-I release [32], [33]. The downloader verified 17 of 17 repository files and streamed 249,316,543 source rows. Preprocessing retained 819,528 observations for Boulder County and its six adjacent counties, including 171,187 Boulder observations. EAGLE-I timestamps are UTC and were converted to America/Denver only for alignment with local charging records. Because the source omits zero-outage records and cannot distinguish a true zero from a collection gap, absent county–time pairs were preserved as missing rather than silently imputed as zero. Events were evaluated at 50, 100, and 500 customers out. At the 50-customer threshold, 2,535 events had median duration 1.75 h and 90th-percentile duration 6.5 h;

TABLE II
FIXED-SPLIT 2023 TEST PERFORMANCE ON MEASURED BOULDER CHARGING SESSIONS

Method	Arrivals		Energy (kWh)	
	MAE	RMSE	MAE	RMSE
Historical mean	0.137	0.517	1.101	3.026
Weekly seasonal	0.156	0.637	1.130	3.299
Persistence	0.175	0.678	1.059	3.186
Graph-recurrent forecaster	0.110	0.527	0.945	3.113

the post-peak half-recovery time was observable for 19.6% of events and had median 0.75 h. These traces inform a scenario distribution, not station-level failure labels or utility work orders.

The independent electrical test uses one complete SMART-DS v1.0 San Francisco P1U feeder and its unmodified OpenDSS files [34], [35]. The solved circuit contains 80 buses, 219 nodes, 27 loads, 67 lines, and 13 transformers. Eighteen nodes already de-energized in the released base case are identified and excluded from active-voltage minima; they are not counted as EV-induced failures. The base active-node voltage minimum is 0.989 p.u. and maximum line loading is 0.913 p.u. Requested EV actions are aggregated as balanced three-phase, wye-connected constant-PQ increments at the 17 eligible three-phase loads. Active power is the executed station request and reactive power follows a 0.98 lagging power factor; the released load voltage level (0.48 or 12.47 kV) and base-load parameters are otherwise unchanged. Twenty seeded station-to-load mappings are prepared in advance; mapping $s \bmod 20$ is shared by all six policies in scenario s and recorded in the scenario table. At every simulated hour, station-level executable charging is aggregated at the assigned loads, OpenDSS solves the unprojected action, and a bisection backtracks its common fraction until voltage lies in $[0.95, 1.05]$ p.u., line loading is at most 1.0 p.u., and the circuit converges. Only this projected energy enters served-energy and backlog equations. This gate validates the electrical feasibility of a proposed charging action. It does not simulate feeder topology reconfiguration, switching or breaker operations, fault isolation, feeder resupply, or equipment-parameter changes after repair; the experiment is therefore a service-restoration benchmark, not a validated feeder switching schedule.

B. Demand forecasting and spatial generalization

Table II reports the fixed 2023 temporal test. The graph-recurrent forecaster achieved arrival MAE 0.110 and energy MAE 0.945 kWh, compared with 0.137 and 1.101 kWh for the historical mean. Its energy RMSE (3.113 kWh) did not beat the historical mean (3.026 kWh), so no uniform metric dominance is claimed. Across three seeds, arrival MAE ranged from 0.1096 to 0.1099 and energy MAE from 0.9452 to 0.9468 kWh.

The inductive spatial test exposed a second boundary. Pooling all five held-out blocks and six horizons, the spatial ridge model reduced energy MAE from 1.130 to 1.054 kWh and RMSE from 2.928 to 2.272 kWh relative to station-wise weekly history. Arrival RMSE also decreased from 0.557 to 0.440, but arrival MAE (0.161) was worse than the pooled calendar mean (0.139). Thus the measured-data forecast supports temporal deployment and improves several unseen-station endpoints, but sparse stations remain a failure mode rather than a hidden success.

TABLE III
PROPAGATION COEFFICIENTS AND PRESPECIFIED UNCERTAINTY SET

Parameter	Central	Interval	Evidence class
ρ_r	0.728	[0.546, 0.911]	estimated
σ	0.233	[0.116, 0.349]	estimated
ρ_p	0.772	[0.754, 0.787]	estimated
τ_{pr}	0.000	[0.000, 0.032]	estimated observational
τ_{pc}	0.166	[0.012, 0.271]	constructed
ρ_c	0.500	[0.300, 0.800]	sensitivity only
τ_{rc}	0.040	[0.000, 0.080]	sensitivity only
τ_{cr}	0.080	[0.000, 0.160]	sensitivity only
τ_{cp}	0.040	[0.000, 0.080]	sensitivity only

C. Propagation calibration and policy comparison

The coefficient ledger in Table III distinguishes estimated, observational, constructed, and sensitivity-only quantities. A nonnegative station-deficit regression estimated road/mobility persistence and spatial spread from 1,268,661 training samples. On 3,108 aligned 2023 test hours, its road-state RMSE was 0.306 versus 0.374 for persistence. EAGLE-I estimated power persistence at 0.772 with a day-block bootstrap interval $[0.754, 0.787]$. The aligned power-to-EV-deficit coefficient was effectively zero with upper interval 0.032, providing no evidence for a strong direct conditional association. The packet module constructed the power-to-communication interval. Directions without joint field traces span sign-constrained sensitivity ranges including zero.

The primary central-PC policy evaluates multistep propagation under the calibrated central matrix. The robust extension evaluates 21 central/endpoint matrices and allocates using the smallest predicted marginal action benefit. In the realized dynamics, each coefficient was sampled independently from its prespecified interval for each scenario. Forecast-matched, central PC, and robust PC received the same selected demand forecast, current threat observation, backlog, and realized past. Static instead used historical station means, greedy used the common forecast but formed a direct current-threat route, and the oracle alone received future innovations. The forecast-matched baseline uses no propagation matrix: it scores the common finite route portfolio from current threat and forecast exposure only. Central PC propagates the calibrated central matrix, whereas robust PC minimizes the worst score over the 21-matrix set. Thus forecast-matched is not central PC. Table IV fixes the information and execution contract for all six policies; the oracle alone receives future innovations. The primary family comprised central PC versus the matched baseline for total cost overall and in four prespecified threat classes. We used 4,096 paired scenarios (1,024 per class), 20,000-resample paired bootstrap intervals, paired t tests, two-sided Wilcoxon tests, and Holm correction across the five classwise tests. The complete parameter ledger is supplied as Supplementary Table S1 and as the machine-readable file `config/full_parameter_ledger.csv`; it reports each value, unit, role, provenance or calibration rule, selection data, and challenged range.

Central PC reduced mean total cost by 0.6327% overall relative to the matched baseline (paired difference -6.2730, 95% bootstrap CI $[-7.0372, -5.5195]$). The Holm-adjusted paired t and Wilcoxon probabilities were 3.84×10^{-56} and 9.76×10^{-101}, respectively. All four prespecified groups had negative intervals, with reductions of 0.0020% in nominal, 0.0433% in single-domain, 0.6502% in cascade, and 1.1296% in OOD compound cases. The nominal effect is statistically distinguishable in these simulated pairs but operationally negligible; the intervals quantify Monte Carlo scenario uncertainty, not variation across real utility systems.

The robust extension had higher mean cost than central PC (986.85 versus 985.17): the robust-minus-central difference was 1.6830 (95% CI $[1.3298, 2.0508]$), or a 0.1708% penalty. Its 95th-percentile cost was equal to central PC (2381.08), its 99th percentile was 7.47 lower, but its CVaR at 95% and 99% was 4.48 and 3.19 higher, respectively, and the maximum was equal (4084.41). Robust PC won 3.83%, tied 88.50%, and lost 7.67% of paired scenarios. These diagnostics do not support either an average-cost or a coherent tail-risk advantage for worst-set scoring, so robust PC is retained only as a sensitivity extension. Central PC reduced cost by 7.885% relative to static allocation. The oracle alone received future innovations; its mean cost was 896.01, leaving central PC with a 9.95% deployable-policy-to-oracle gap. The oracle is therefore an information-value reference, not evidence of near-optimality.

Reweighting one objective component at a time did not reverse the central-PC comparison with forecast-matched: the reduction ranged from 0.4105% to 0.7978%, and every paired bootstrap interval remained below zero. These fixed- trajectory tests assess evaluation-weight dependence; they do not claim that a policy re-optimized for each alternative objective would select the same route.

D. Action-in-loop electrical and mobility feasibility

The primary 4,096-scenario run made 147,456 online OpenDSS calls, one for each scenario–policy–hour. At measured load, all raw actions were already feasible, so projection retained 100% of requested charging; this is a valid feasibility result but cannot by itself demonstrate feedback. The companion $20\times$ stress run therefore used 1,024 paired scenarios and the same loop. It produced 1,089 raw-infeasible policy-hours; projection reduced this count to zero, retained 99.26% of requested service on average, and placed 69,664.8 kWh of curtailed energy into unserved demand/backlog rather than crediting it as served. Thus the online adapter changes downstream state whenever the AC limit binds.

The SMART-DS experiment comprised 20 randomized station-to-load mappings, 24 test hours, and four EV load multipliers (1,920 cases). Raw actions were all feasible at measured load. At $3\times$, $5\times$, and $10\times$ load, raw feasibility fell to 83.1%, 44.6%, and 25.4%, whereas the action projection retained 98.2%, 81.1%, and 57.7% of requested EV power. Every projected case converged and met the voltage and line limits. A separate full-2023 test used all 7,451 nonzero hours at $1\times$ and $3\times$ load (14,902 cases). At $3\times$, 99.17% of raw actions were feasible and projection retained 99.94% on average; all projected cases remained AC-feasible. The reported curtailment is therefore produced inside the action path and not inferred from a post-hoc voltage label.

Station closure does not automatically imply abandoned charging. A separate capacity-constrained flow model distinguished local service, rerouting, and unserved demand across 24 observed peak hours. Under measured demand and 10, 20, and 30% closures, preserving the original station served only 67.4, 55.8, and 43.7% of energy. Nearest, logit, and coordinated reassignment all served 100% in these cases. The coordinated solution required only 0.32, 0.50, and 0.97 km energy-weighted extra distance. Under $4\times$ demand, total constructed capacity, rather than the choice model, became the limiting factor; this stress result prevents the rerouting mechanism from creating fictitious capacity.

E. Packet-level control feedback and executable crew routes

The packet emulator is a finite-buffer first-come, first-served discrete-event queue with erasures, two retransmissions, backoff, a 500 ms control deadline, and explicit backup duration. An M/M/1 unit test at 10 packet/s arrival and 20 packet/s service reproduced the theoretical 100 ms mean latency with 0.269% relative error. Across 12,000 station stress cases, the original algebraic support proxy had Spearman correlation 0.860 but mean absolute error 0.243 against the simulated on-time action fraction, which confirms that the proxy cannot replace packet dynamics.

Equation (8) closes this evidence loop. Under prespecified $2\times$ traffic and 1,024 paired uncertain-transition scenarios, 60 s backup raised the mean on-time action fraction to 0.938 and reduced central-PC cost by 10.54% relative to no backup (difference CI $[-147.3, -126.8]$). A 300 s backup raised the fraction to 0.969 and reduced cost by 18.83% (difference CI $[-261.2, -227.7]$). Before backup exhaustion, power threat caused no direct service-rate derating; the effect arose only after the coded depletion event.

In the integrated primary run, the four-crew route was not a diagnostic added after scoring. The same work-order set was dispatched through packet delivery, and threat was updated only at its recorded completion hours. Central PC completed 82.17% of dispatched jobs versus 81.72% for the forecast-matched baseline, reduced mean job completion time from 3.131 to 3.095 h, and reduced crew travel from 2.975 to 2.957 h per scenario. Paired bootstrap intervals for the differences were $[0.0038, 0.0051]$, $[-0.0417, -0.0300]$ h, and $[-0.0263, -0.0102]$ h, respectively.

The route experiment used 36 jobs, integer fleets of 4, 12, and 24 crews, geocoded travel at 30 km/h with 1.25 circuity, and service-time multipliers 0.5, 1, and 2. Each repair completion causally removed that job's risk from the accumulated state. The original PC priority lost its advantage in some integer routed cases. Route-aware PC corrected this failure by selecting the lowest predicted-risk executable candidate rather than projecting a continuous priority after the decision. In

TABLE IV

POLICY INFORMATION AND EXECUTION CONTRACT. "CURRENT" MEANS THE OBSERVATION AVAILABLE AT THE DECISION HOUR. THE COMMON ROUTE PORTFOLIO CONTAINS ROUTES INDUCED BY STATIC, GREEDY, FORECAST-MATCHED, CENTRAL-PC, ROBUST-PC, AND UNIFORM PRIORITIES. FUTURE INNOVATIONS ARE NEVER REVEALED TO DEPLOYABLE POLICIES.

Policy	Demand information	Threat/backlog	Multi-step transition rollout	Transition matrix used	Future innovations	Repair-route rule	Common execution gates
Static	Historical station mean	Backlog only for charging	No	None	No	Direct route from static priority	Yes
Greedy	Common forecast	Current threat and backlog	No	None	No	Direct route from current-threat priority	Yes
Forecast-matched	Common forecast	Current threat and backlog	No	None	No	Current-exposure score over common portfolio	Yes
Central PC	Common forecast	Current threat and backlog	Yes	Calibrated central M	No	Central-transition score over common portfolio	Yes
Robust PC (extension)	Common forecast	Current threat and backlog	Yes	Worst score over 21 declared matrices	No	Worst-transition score over common portfolio	Yes
Oracle	Realized future demand	Current threat and backlog	Yes	Central M	Yes	Direct route from oracle priority	Yes

TABLE V

PRIMARY TRANSITION-AWARE COMPARISON AND ALL-POLICY COST SUMMARY UNDER SAMPLED TRANSITION UNCERTAINTY. POSITIVE REDUCTION DENOTES LOWER COST THAN FORECAST-MATCHED; THE ORACLE ALONE RECEIVES FUTURE DEMAND AND INNOVATIONS.

Threat group	Pairs	Central PC	Matched baseline	Reduction (%)	Difference 95% CI
All	4096	985.17	991.44	0.6327	[-7.0372, -5.5195]
Nominal	1024	661.82	661.83	0.0020	[-0.0233, -0.0053]
Single domain	1024	689.06	689.35	0.0433	[-0.4778, -0.1179]
Cascade	1024	985.30	991.75	0.6502	[-7.5902, -5.3153]
OOD compound	1024	1604.49	1622.82	1.1296	[-21.0397, -15.6852]

All-policy mean over the same 4,096 scenarios

Policy	Mean cost	Reduction vs. matched (%)
Static	1069.50	-7.873
Greedy	992.07	-0.063
Forecast-matched	991.44	0.000
Central PC	985.17	0.633
Robust PC	986.85	0.463
Oracle	896.01	9.626

TABLE VI

CENTRAL-PC EVALUATION-OBJECTIVE SENSITIVITY RELATIVE TO FORECAST-MATCHED ON THE SAME 4,096 PAIRED TRAJECTORIES. POLICIES ARE NOT RE-OPTIMIZED UNDER ALTERNATIVE WEIGHTS.

Evaluation weights	Reduction (%)	Paired difference (95% CI)
Reference weights	0.6327	-6.2730 [-7.0372, -5.5195]
2× mobility weight	0.6375	-6.3821 [-7.1811, -5.6118]
2× unserved-energy weight	0.4105	-6.9247 [-7.7991, -6.0867]
2× communication weight	0.7978	-8.9713 [-10.0243, -7.9314]
2× power-service weight	0.6411	-6.5376 [-7.4521, -5.6625]
0.5× cascade weight	0.5380	-4.9984 [-5.6519, -4.3621]

128 independently seeded scenarios per cell, route-aware PC reduced mean integrated risk in all nine crew/service conditions by 0.31–1.08%. Six bootstrap intervals excluded zero and seven Holm-adjusted Wilcoxon tests were below 0.05. The 4-crew/1.0, 12-crew/1.0, and 24-crew/0.5 intervals included zero and remain visible in Table XI; the implementation gain is not declared uniformly significant. The 24-crew/0.5 Holm-adjusted Wilcoxon value is below 0.05 while its bootstrap interval crosses zero because the two procedures test different statistics: the interval targets the paired mean difference, whereas Wilcoxon ranks the nonzero paired differences and is affected differently by ties and skewness. We therefore require the mean interval, not either test in isolation, for a mean-improvement claim.

The integrated-loop sensitivity runs used another 1,024 paired scenarios per condition. Central PC reduced cost relative to matched by 0.529%, 0.421%, and 0.668% under $0.75M$, $1.25M$, and 0.15 coefficient noise, respectively. The robust-minus-central difference was -0.470 at $0.75M$ (95% CI $[-1.034, 0.028]$), but $+6.383$ at $1.25M$ (CI $[4.910, 7.976]$) and $+2.272$ under coefficient noise (CI $[1.547, 3.069]$). Thus the only favorable overall robust comparison was not statistically distinguishable from zero, whereas two misspecification settings favored central PC. With 12 crews robust PC was 0.075 cost units worse than central PC (CI $[0.005, 0.165]$); with 24 crews its 0.022-unit advantage was negligible and the interval crossed zero.

F. Implementation and proof checks

The source-to-theorem audit executed 11 independent checks. Ten thousand random score/budget tests preserved each continuous budget (maximum numerical error 1.37×10^{-4}); 5,000 random largest-remainder tests preserved the integer crew count; all 8,117 nonzero 2023 test hours satisfied the analytical zero-service backlog bound; 100,000 steps at simultaneous upper uncertainty bounds remained inside $[0, 0.98]^{3N}$; all 14,902 standalone SMART-DS projections,

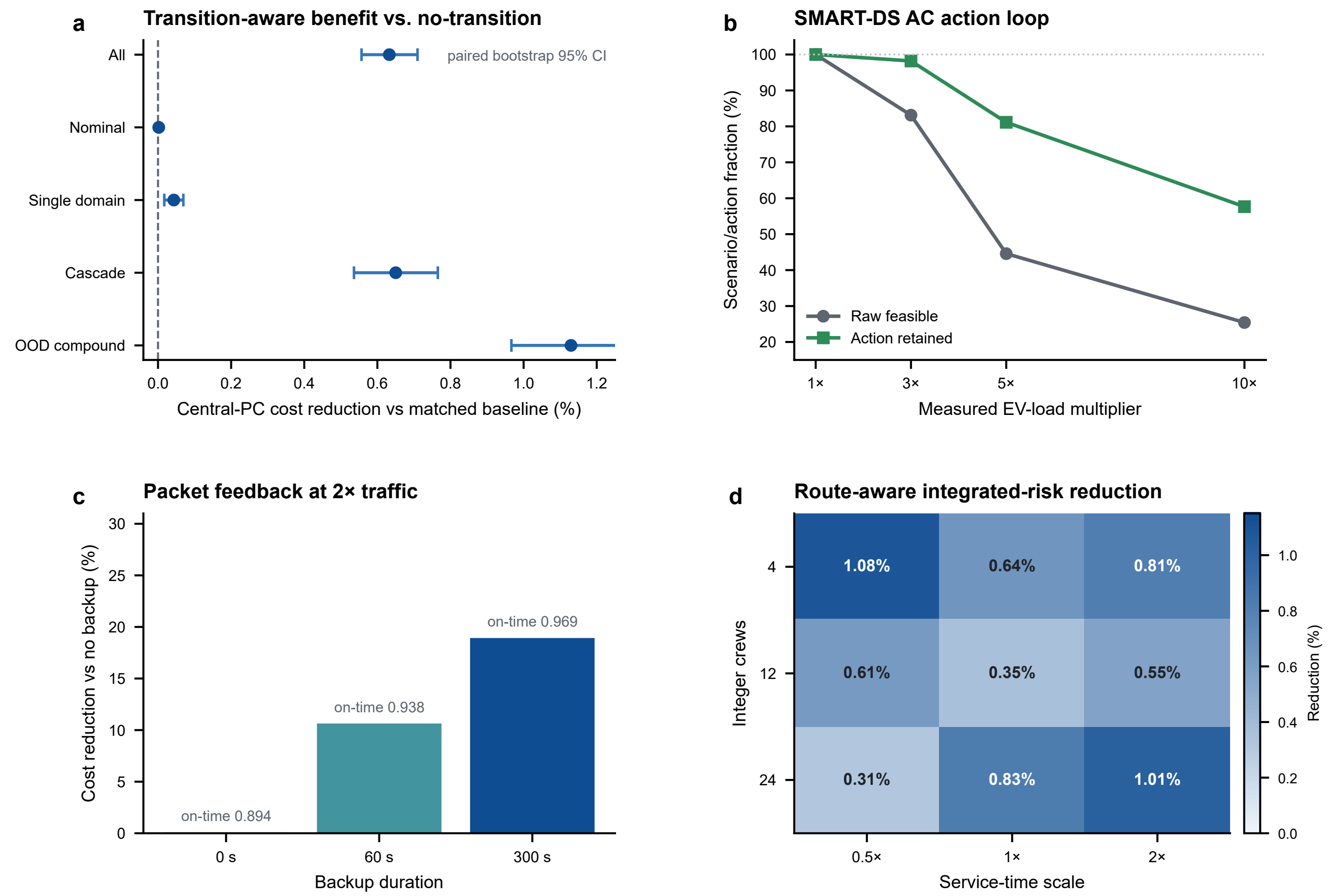

Fig. 2. Paired transition-aware and execution evidence. (a) Central-PC cost reduction relative to the information-matched no-transition baseline; points are paired means and bars are 20,000-resample bootstrap 95% intervals ($n = 1,024$ per threat class and $n = 4,096$ overall). (b) Raw AC feasibility and the fraction of requested EV action retained by SMART-DS projection across 20 mappings and 24 hours per load multiplier. (c) Cost reduction relative to no backup at 2× packet traffic; annotations give mean on-time action fraction ($n = 1,024$ paired scenarios). (d) Route-aware PC integrated-risk reduction for 128 paired scenarios per integer-crew/service-time cell. Holm-adjusted tests and exact intervals for routing are reported in Table XI. Source data are supplied with the figure.

TABLE VII
SMART-DS PROJECTION AND ROUTED CREW EVENTS INSIDE THE CLOSED-LOOP EVALUATION

Condition	Pairs	Online AC hours	Raw infeasible	Projected infeasible	Action retained (%)	Crew completed (%)
Measured-load primary	4,096	147,456	0	0	100.00	81.86
20× EV execution stress	1,024	36,864	1,089	0	99.26	81.92

TABLE VIII
SMART-DS AC ACTION PROJECTION (20 MAPPINGS, 24 HOURS EACH)

EV load	Raw feasible (%)	Action retained (%)	Raw $V_{\min}$	Raw line p.u.
1×	100.0	100.0	0.986	0.929
3×	83.1	98.2	0.977	0.968
5×	44.6	81.1	0.966	1.044
10×	25.4	57.7	0.938	1.361

TABLE IX
CROSS-STATION CHOICE UNDER MEASURED DEMAND (24 PEAK HOURS)

Closure (%)	No choice	Nearest	Logit	Coordinated	Extra km
10	67.4	100.0	100.0	100.0	0.32
20	55.8	100.0	100.0	100.0	0.50
30	43.7	100.0	100.0	100.0	0.97

147,456 primary online calls, and 36,864 stress online calls were feasible after projection; packet backup and action-derating assertions had zero logic error; every audited route visited each selected job once; all main crew counts matched completion-event logs with paired inputs; and all 2,592 station-choice flows met capacity. The machine-readable map links each property to its theorem, function, evidence file, and status.

VI. Discussion

The revised evidence changes the interpretation of PC rollout. The central advance is not a large average gain from adding another forecast module. It is an auditable procedure

TABLE X
BACKUP-POWER EFFECT WITH PACKET FEEDBACK AT $2\times$ TRAFFIC ($n = 1024$ PAIRED SCENARIOS)

Backup	On-time action	Restoration enacted	Cost reduction (%)	Difference 95% CI
60 s	0.938	0.756	10.54	[-147.3, -126.8]
300 s	0.969	0.789	18.83	[-261.2, -227.7]

TABLE XI
ROUTE-AWARE PC VERSUS MATCHED ROUTING BASELINE

Crews	Service scale	Risk reduction (%)	Difference 95% CI	Holm p
4	0.5	1.08	[-2.25, -0.47]	0.0237
4	1.0	0.64	[-2.33, 0.06]	0.177
4	2.0	0.81	[-2.85, -0.94]	0.00175
12	0.5	0.61	[-0.65, -0.12]	0.0492
12	1.0	0.35	[-0.68, 0.00]	0.129
12	2.0	0.55	[-1.63, -0.20]	0.00652
24	0.5	0.31	[-0.30, 0.01]	0.0293
24	1.0	0.83	[-0.94, -0.37]	6.49e-05
24	2.0	1.01	[-1.95, -0.82]	6.91e-06

that preserves a modest transition-aware benefit over a no-transition, information-matched ablation and that exposes how electrical, communication, station-choice, and integer-routing constraints change the action before it is credited. The online primary and stress runs show this inside the same implementation rather than through independent post-hoc validators. The strong backup-power effect is a design result of the packet/control loop, whereas the small nominal PC advantage shows that multi-step propagation is not valuable in every operating state. Because central PC had 0.1708% lower mean cost than robust PC and the robust extension did not improve CVaR or maximum cost, the evidence does not establish a worst-set protection benefit.

Several alternative explanations were tested directly. Forecast performance alone cannot explain the central-PC–forecast-matched comparison because those two policies receive the same forecast. Separating SMART-DS from policy scoring substantially mitigates, but does not eliminate, circular validation: the independent solver enforces AC constraints before served energy updates backlog, yet the relative policy benefit is still estimated with the same service-loss primitives and a modeled transition environment. Moreover, the measured-load AC gate was nonbinding for every raw primary action, so its role there was independent feasibility verification rather than a change in policy ranking; the $20\times$ stress experiment demonstrates feedback only when the constraint binds. We therefore report evaluation-weight sensitivity and objective-external operational endpoints (AC feasibility, retained charging, crew completion time, and travel time) separately from total cost. The crew result is not a continuous-effort artifact because repair occurs only at logged integer-route completion events. Conversely, the failure of the graph predictor to dominate every RMSE and the original PC priority's loss under routing are retained as negative results; both motivated concrete model changes and constrain the claim.

The remaining boundaries are data-specific rather than deferred substitutes for validation. Boulder session energy is measured, but the intra-session hourly profile is reconstructed. EAGLE-I is county-level and has ambiguous missing intervals, so it cannot identify station outages or causal crew times. SMART-DS is a realistic synthetic feeder, not the Boulder utility network. The communication traffic and battery conditions are transparent stress inputs, not a field packet trace. These boundaries prevent a utility-deployment claim, but they do not remove the measured demand test, held-out outage calibration, independent AC action loop, packet feedback, or executable routing evidence on which the stated benchmark conclusion rests.

VII. DATA AND CODE AVAILABILITY

The public repository and accompanying reviewer-minimal archive provide the code, processed inputs, scenario-level outputs, and selected SMART-DS/OpenDSS feeder needed to reproduce the reported principal statistics and executable checks. The original Boulder CSV and 17 EAGLE-I files remain available from their public publishers and are not duplicated in the minimal archive; retained metadata record their source URLs, licenses, access dates, byte counts, and SHA-256 hashes. Processed tables preserve data dictionaries, split boundaries, missing-value semantics, timezone decisions, and conservation checks. Pinned Python dependencies, random seeds, trained weights, scenario-level outputs, statistical tables, 20 station-to-SMART-DS mappings, hourly AC projection aggregates, integer route/completion logs, packet logs, proof checks, the integrated $20\times$ stress run, and one-command verification are included. A standalone complete-parameter supplement and its machine-readable CSV avoid requiring reviewers to inspect the full archive merely to identify a setting. A reviewer-minimal archive, including a smoke test and the scenario-level data needed to regenerate the principal statistics, accompanies this revision; no restricted data or HPC access is required. The corresponding public repository is available at https://github.com/ahshicr/TSG-01697-2026-reproducibility (tagged release `v2026.09.01`). The supplied files are versioned by a SHA-256 manifest and contain a self-checking minimal reproduction path. Publicly reused datasets retain their original source URLs, licences, versions, and identifiers. No separate repository DOI is required or claimed.

VIII. CONCLUSION

Transition-aware PC rollout combines measured charging demand, an action-dependent calibrated transition, and executable electrical, communication, and crew actions. Across 4,096 paired scenarios central PC reduced mean cost by 0.6327% over the information-matched no-transition ablation, with larger effects in cascade and OOD groups and the same direction under prespecified objective reweighting. A 21-matrix worst-set extension was 0.1708% worse in mean cost and improved neither CVaR nor maximum cost; it is therefore a sensitivity variant rather than the principal method. Every hourly primary action was processed by SMART-DS and integer crew completion events inside that same loop; a companion electrical stress test verified that binding AC constraints change served energy and backlog. Packet, station-choice, and standalone route experiments further identified the constraints that materially change implementation. Backup power produced a large control-loop benefit under communication stress, whereas mild-operation rollout gains were negligible and unseen-station errors remained metric-dependent. The theory and property checks establish finite-horizon sensitivity for the continuous predictive operators, a conditional finite-portfolio regret bound, and separate constraint-preservation results for execution. The evidence supports a reproducible public-data restoration benchmark; county-level outage context, a synthetic independent feeder, and constructed packet traffic do not support a claim of utility field deployment.

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